VishingDetector

By: Sarah Ma, Justin Tam, Jack McSherry

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OUR TEAM & MOTIVATION

Justin Tam
Undergrad CS @ USF

Jack McSherry
Undergrad CS @ USF

Sarah Ma
Undergrad CS @ USF

- AI generated media is getting harder to discern from human media.
- Voice cloning now only needs a few seconds of reference audio.
- The quality of AI generated voice is increasing at a rate close to AI development.
- A Harvard/Meta study found 36% compliance when a “relative” was in distress.

DESCRIPTION & SCOPE

PROBLEM

Threat: Voice phishing using voice-cloning and AI speech

Objective: Build a model to detect these attacks

SCOPE

Covers both generic AI-generated voices and targeted clones

- **Input**: Audio file
- **Output**: Binary classification + confidence percentage

Significance: As AI improves, voice synthesis quality outpaces traditional verification

Applications:

- Corporate defense
- Banking/identity verification
- Personal protection

BACKGROUND

graystafford/deepfake-audio-detection

Data

Self-collected dataset:

- 933 samples from 14 YT recordings
- 933 generated samples
- **Sample lengths**: 2.5-13 seconds

Approach

- **Feature Extraction**: Wav2Vec2 extracts acoustic features.
- **Encoding**: Transformer layers encode feature representations.
- **Classification**: Linear Classification with Softmax output.

Source: https://huggingface.co/garystafford/wav2vec2-deepfake-voice-detector

anantteo/deepfake-audio-detection

Data

- 20 Speakers (8 Men, 12 Women)
- 2 voice conversions, 4 speech synthesis algorithms
- **Sample lengths:** 1-10 seconds

Approach

- **Transformers:** Wav2Vec2, HuBERT, AST
- **Baselines**: CNN, MLP, Random Forest, SVM.

Source: https://github.com/anantteotia/Audio-Deepfake-Detection

DATA

ASVspoof 2019 Logical Access (LA) and ASVspoof 2021 Evaluation datasets.

Train & Tune Split:

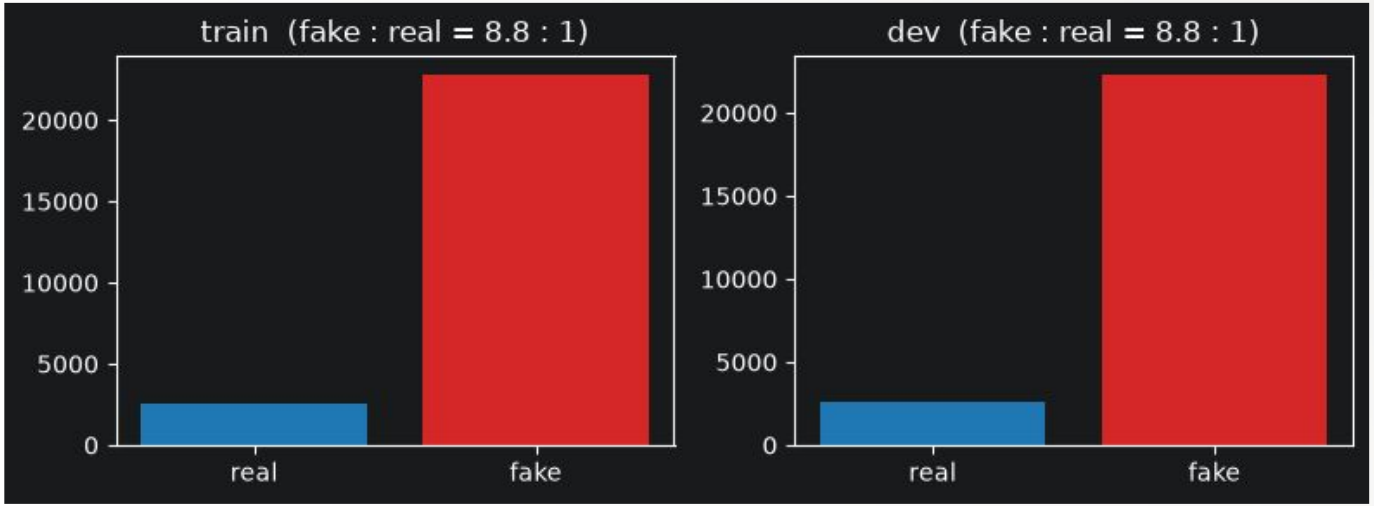
Train: 2019 LA training set (25,380 clips; 6 fake-making methods)

Tune: 2019 LA development set (24,844 clips)

Final Test Set: 2021 LA and Deepfake (DF) evaluation sets (unseen attack types, phone/compression effects)

Class Imbalance

Handling Class Imbalance: Fakes outnumber real clips (~9:1 ratio); using class weighting and Equal Error Rate (EER) rather than raw accuracy.



FEATURES

Spectral & Cepstral Features

MFCCs & LFCCs

Log-mel & CQT Spectrograms

Voice-Production & Artifact Clues

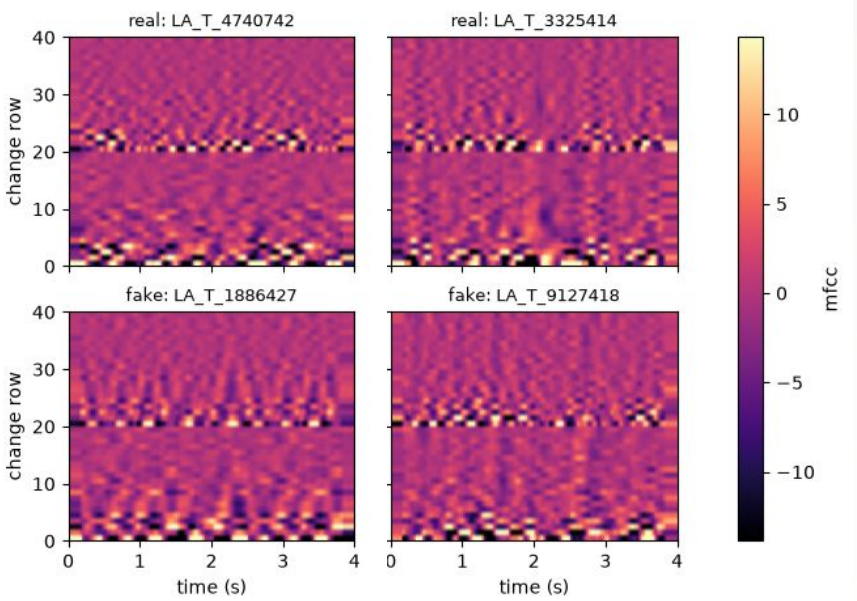
Pitch Statistics

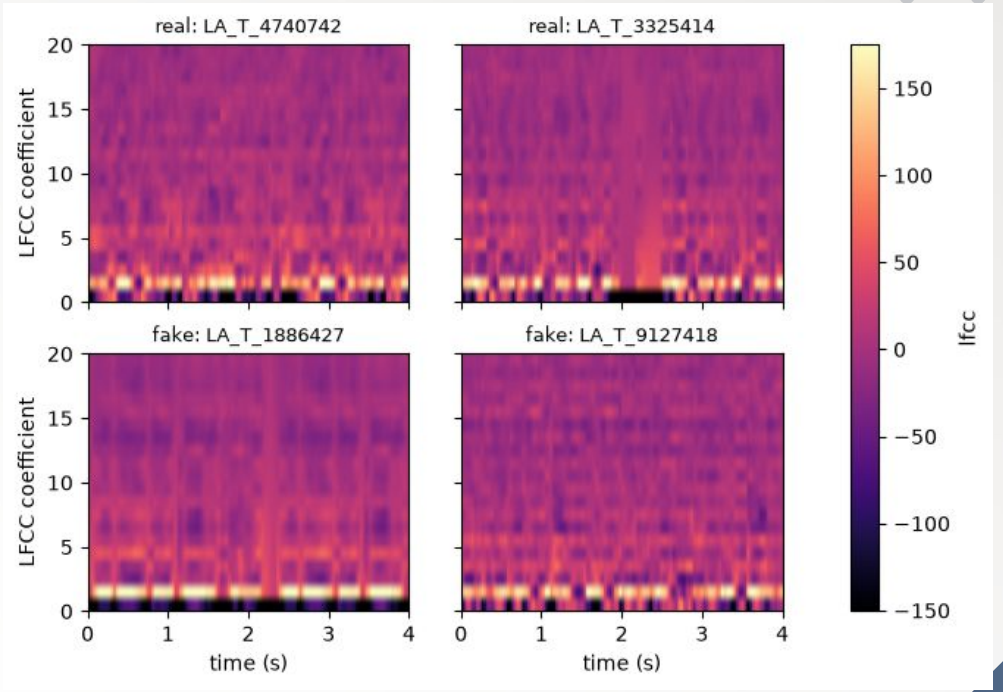
Jitter & Shimmer

Background Noise

FEATURES – MFCC & LFCC

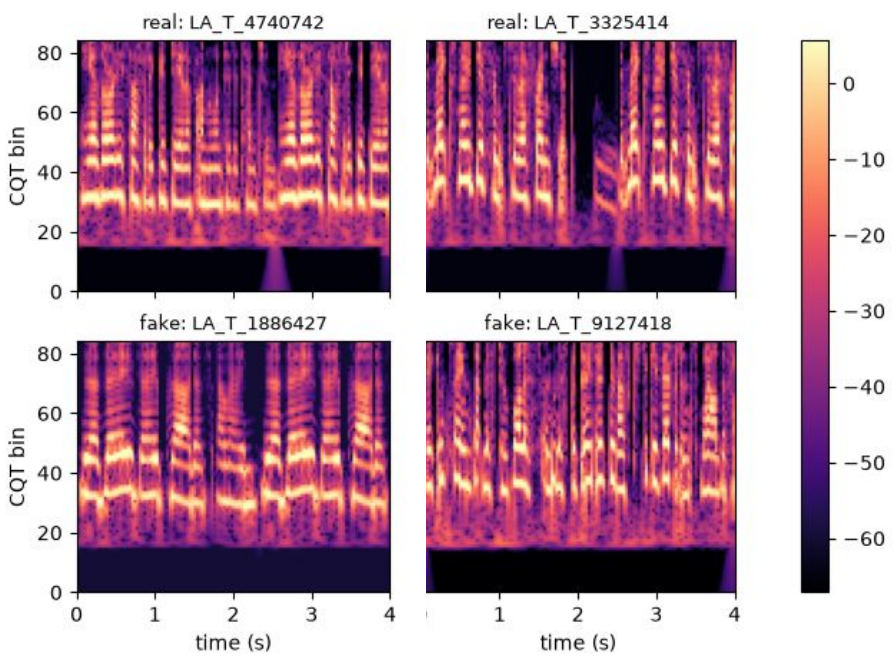
MFCCs & LFCCs: Capturing voice shape and high-frequency pitch details along with delta/delta-delta changes.

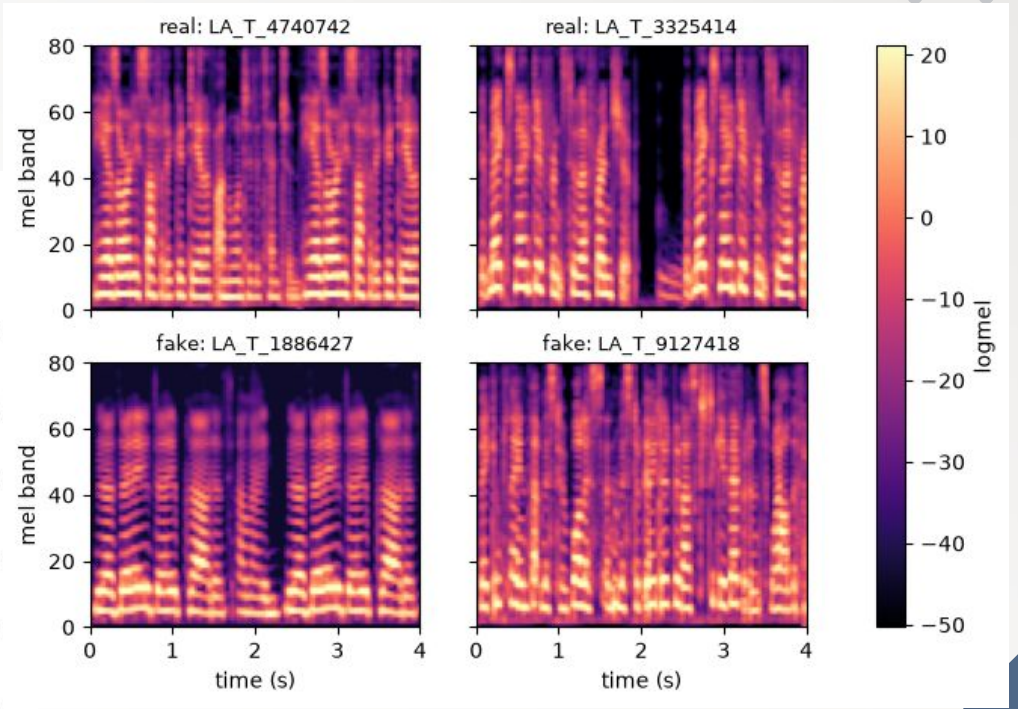




FEATURES - Log-mel & CQT

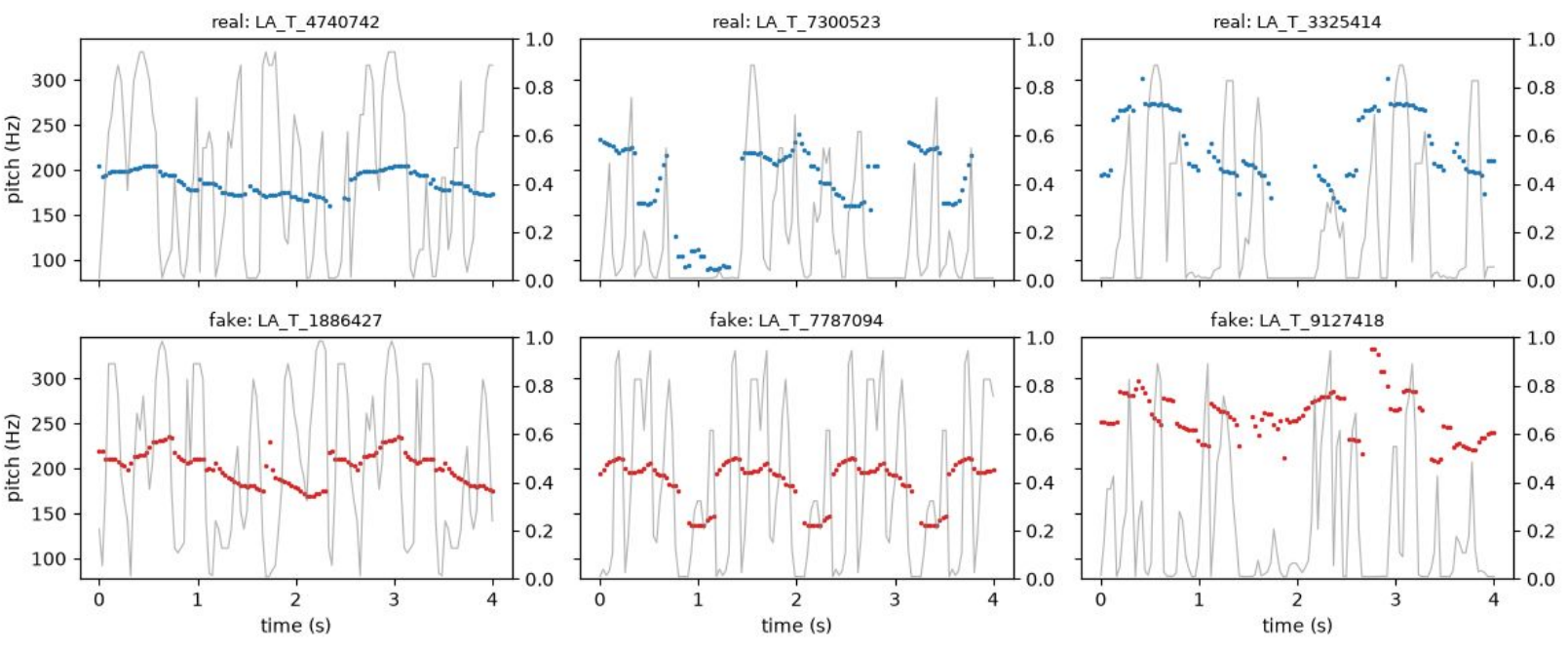
Spectrograms & CQT: Mapping time-frequency energy distribution and constant-Q pitch ranges.





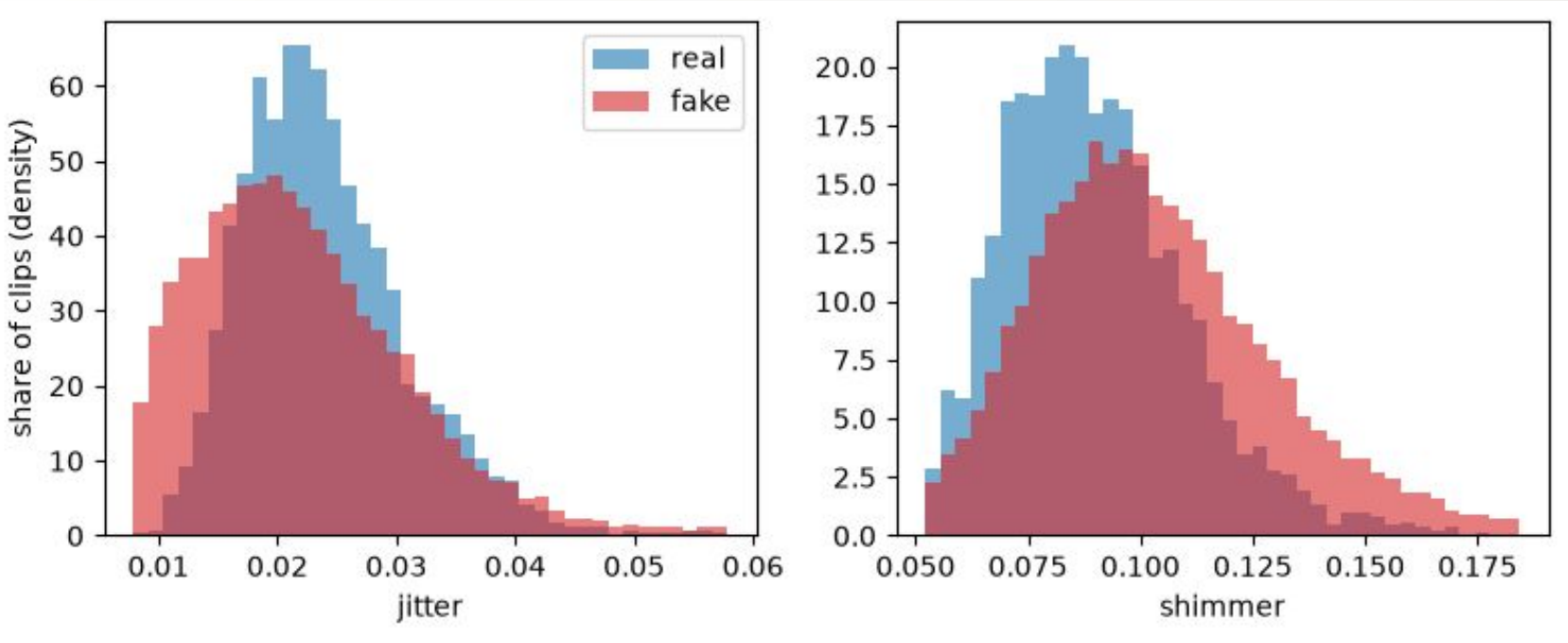
FEATURES – Pitch Statistics

Pitch Statistics: Tracking fundamental frequency (f_0) mean,, standard deviation, and transitions.



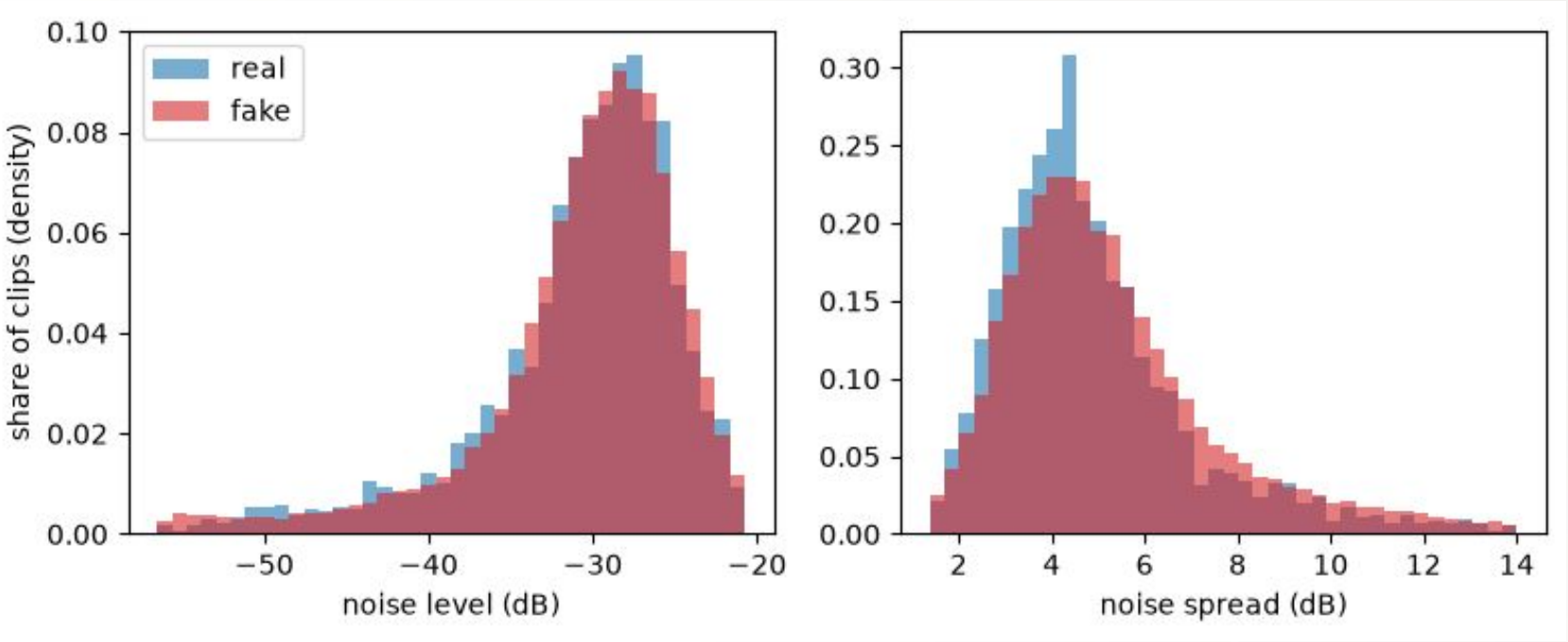
FEATURES - Jitter & Shimmer

Jitter & Shimmer: Measuring micro-instabilities in pitch and loudness to identify synthetic wobbles.



FEATURES - Noise

Background Noise & Silence: Analyzing quiet intervals and RMS energy distributions to flag unnaturally clean or poorly compressed audio.



Testing on other data

graystafford/deepfake-audio-detection

Data

Real: 44.1 kHz stereo
Fake: 16 kHz mono

Real: 993 samples
Fake: 933 samples

Takeaways

Best features: CQT and LFCC

Weight of the features changes when grouped together, implications for the main dataset

Thank you!